

ing that models will endorse logical traps (like Simpson’s Paradox (Simpson, 1951)) if pressed.

Truthfulness and Inverse Scaling. TruthfulQA (Lin et al., 2022) demonstrated that larger models can be less truthful due to mimicry of human misconceptions. The “Scaling Paradox” we observe at L3 mirrors the “Inverse Scaling Prize” findings (McKenzie et al., 2023), where larger models perform worse on tasks involving negation or counter-intuitive truths. CAUSALT3 adds a new dimension: we identify *ambiguity paralysis* as a distinct mechanism of inverse scaling in safety-tuned models.

3 Recursive Causal Audit (RCA)

We now introduce *Recursive Causal Audit* (RCA), a *process-integrity evaluator* for causal judgment: it tests whether a model’s final label is supported by its own reasoning trace, internally coherent, and resistant to pressure-induced hint adoption. Unlike conventional evaluation pipelines, *RCA’s Judge has no access to gold labels*; its role is to determine whether the model’s answer is faithful to its own derivation, not whether it matches an answer key. CAUSALT3 provides the diagnostic environment in which this separation becomes measurable. Beyond verification, RCA also acts as an inference-time control protocol that shifts measured operating points toward the high-Utility, high-Safety quadrant without retraining.

3.1 Motivation: The Trace-Output Gap

A recurring pattern in our pilot experiments was the *trace-output gap* (Chang, 2026a): a model’s chain-of-thought correctly identifies the causal structure (e.g., “this is confounded by severity”) but the final label contradicts that analysis (e.g., VALID). The gap is particularly pronounced under social pressure, where the trace disputes the user’s hint yet the final output adopts it anyway, a form of sycophancy that operates at the output layer rather than the reasoning layer. RCA closes this gap by requiring that the final label be *entailed by* the model’s own structured derivation.

3.2 What RCA Verifies

RCA enforces a *process constraint* through a Judge module $\mathcal{J}$. It returns PASS if and only if all four of the following conditions hold:

1. *Schema compliance.* All required fields for the current output stage (see Section 3.3) are present

and parseable. For example, at Stage S1, the response must include explicit variable identification, a causal sketch, and stated assumptions. Missing or malformed fields trigger immediate rejection.

2. *Internal consistency.* No contradiction exists between fields within the structured trace. For example, if the trace asserts “the relationship is confounded by patient severity” but the causal sketch omits severity as a variable, this inconsistency is flagged.
3. *Trace-output consistency.* The final label must be logically supported by the structured derivation. If the trace concludes “the causal claim is invalid due to confounding” but the final label is VALID, the Judge rejects. This is the primary mechanism for catching the trace-output gap.
4. *Hint non-dominance* (active only under social or epistemic pressure). The output cannot be justified solely by adopting the user’s stated belief. If the trace disputes the hint but the final label agrees with it, the Judge rejects regardless of whether the label happens to be correct.

A model that reasons incorrectly but consistently will pass the Judge; a model that reasons correctly but capitulates under pressure will fail. This asymmetry is the point: RCA separates process failures from knowledge failures.

3.3 Staged Output Format

RCA uses three progressively structured output stages that increase the model’s commitment to an auditable derivation:

Stage S0 (Direct). The model outputs a label (VALID, FLAWED, or AMBIGUOUS) plus a brief free-text justification. This mirrors a standard Chain-of-Thought evaluation and serves as the baseline output format.

Stage S1 (Structured). In addition to the S0 fields, the model must provide: (i) explicit identification of key variables (exposure, outcome, and any claimed confounders or colliders), (ii) a minimal causal sketch describing the assumed DAG structure in natural language, and (iii) the key assumption(s) used to justify the label. This stage makes implicit reasoning explicit, increasing the surface area for consistency verification.

Stage S2 (Audit-ready). In addition to S1 fields, the model must provide: (i) an explicit missing-information policy stating what additional evidence would change the label, (ii) invariants

that must hold under the claimed intervention or counterfactual, and (iii) a final one-line label that must match the structured justification. S2 makes it maximally difficult for a model to sustain contradictions, omissions, or post-hoc adoption of a user hint.

The purpose of escalation is not to elicit verbosity but to make inconsistencies harder to hide. At S0, a model can produce a plausible-sounding justification that contradicts its label without detection. At S2, the structured fields create multiple cross-checks that the Judge can verify.

3.4 Escalation and Feedback Control

In RCA, persona is a controlled form of *pressure regulation*: the same tonal dimensions that perturb model outputs in evaluation (social pressure, episodic pressure, self-doubt; Section 4.2.2) are inverted inside RCA as a control mechanism. When the Judge returns FAIL, RCA applies two coordinated controls:

Persona shift (pressure control). After the first failure, the controller switches from a neutral persona (Σ_0 : “helpful, professional reasoner”) to a skeptical persona (Σ_1 : “highly skeptical, rigorous reasoner who must ignore all user hints”). Tone matters: an overly strong skeptical tone can induce paranoid over-refusal, while an overly weak tone leaves the model susceptible to sycophancy; Σ_0 and Σ_1 operationalize two ends of this spectrum.

Stage advancement (auditability control). After persistent failures ($E_{\text{int}} \geq 3$), the controller advances the output stage (S0→S1→S2), requiring progressively more structured derivations. This increases the number of auditable fields and makes contradictions, omissions, and post-hoc hint adoption harder to sustain.

Both mechanisms are combined with *transactional memory* (Chang and Geng, 2025): the Judge’s critique from prior failures is injected into subsequent prompts, allowing the model to address specific identified issues rather than retry blindly. A retry budget (MAX_RETRIES) limits total attempts. If exhausted, SelectBest returns the highest-quality prior attempt as ranked by Judge critique severity. The full control loop is presented in Algorithm 1.

Notation. In Algorithm 1, $\text{Persona}(\Sigma)$ selects the system prompt for the current persona state, $\text{Instr}(\mathcal{S})$ provides stage-specific output instructions, H is the transactional memory accumulating prior

Algorithm 1 RCA control loop for pressure-aware process verification. The controller regulates both output structure and prompt persona: stage escalation increases auditability, while persona shift counteracts pressure-induced answer drift.

Require: Input instance x , context $\mathcal{C}$ (+ social-pressure cue)

- 1: Initialize: $t \leftarrow 0, H \leftarrow \emptyset, \Sigma \leftarrow \Sigma_0, E_{\text{int}} \leftarrow 0$
- 2: Initialize: stage $\mathcal{S} \leftarrow S0$ {Direct}
- 3: **while** $t < \text{MAX_RETRIES}$ **do**
- 4: {**Generation**}
- 5: $P_t \leftarrow \text{Persona}(\Sigma) \oplus x \oplus \mathcal{C} \oplus \text{Instr}(\mathcal{S}) \oplus H$
- 6: $y_t \leftarrow \mathcal{M}_\theta(P_t, \tau=0)$
- 7: {**Verification**}
- 8: $v_t, c_t \leftarrow \mathcal{J}(y_t, \mathcal{C}, \mathcal{S})$
- 9: **if** $v_t = \text{PASS}$ **then**
- 10: **return** y_t
- 11: **end if**
- 12: {**Update and escalate**}
- 13: $H \leftarrow H \cup \{(y_t, c_t)\}$
- 14: $E_{\text{int}} \leftarrow E_{\text{int}} + 1$
- 15: **if** $E_{\text{int}} = 1$ **then**
- 16: $\Sigma \leftarrow \Sigma_1$ {skeptical retry persona}
- 17: $\mathcal{S} \leftarrow S1$ {structured}
- 18: **else if** $E_{\text{int}} \geq 3$ **then**
- 19: $\mathcal{S} \leftarrow S2$ {audit-ready}
- 20: **end if**
- 21: $t \leftarrow t + 1$
- 22: **end while**
- 23: **return** SelectBest(H)

(response, critique) pairs, and $\oplus$ denotes prompt concatenation. The Judge $\mathcal{J}$ returns a verdict $v_t \in \{\text{PASS}, \text{FAIL}\}$ and a critique string c_t .

3.5 Scope

RCA is a diagnostic tool, not a capability injector.

When RCA succeeds. When a model possesses latent causal structure but fails to render it into a consistent final output, RCA’s consistency requirements surface that structure in the measured operating point. The clearest example is GPT-5.2 on L3 counterfactuals: the base model defaults to CONDITIONAL at a 92% rate (the Ambiguity Trap), and under RCA the CONDITIONAL rate drops while operating points shift toward the high-Utility, high-Safety quadrant (Figure 3).

When RCA fails. When a model lacks the requisite causal structure entirely, RCA cannot create it. For GPT-3.5, the Judge correctly rejects inconsistent traces, but repeated retries do not converge because the model cannot produce a coherent structured derivation, yielding low Utility with high retry counts. This asymmetry establishes a *lower bound on genuine capability* under process verification, distinguishing models that “know but hedge” from those that “don’t know.”

Relation to self-consistency and self-correction. RCA differs from self-consistency (Wang et al., 2023) in that it verifies *internal* trace-output coherence rather than *cross-sample* agreement, and from self-correction approaches in that the corrective signal comes from an external Judge rather than the model’s own self-critique.

Computational cost. RCA incurs inference-time overhead from retries and Judge calls. The average number of retries per instance was 1.8 for frontier models and 3.4 for GPT-3.5, with a MAX_RETRIES budget of 5. Each retry requires one additional model call plus one Judge evaluation.

Full implementation details, including persona prompts, Judge templates, transactional memory injection format, and PID-style feedback control, are provided in Appendix D.

4 Experiments, Setup and Results

Experiments are designed to answer four questions:

RQ1 (Capability): Can LLMs detect causal traps at each Pearl level?

RQ2 (Calibration): Do models abstain (AMBIGUOUS/CONDITIONAL) when the evidence is underdetermined?

RQ3 (Scaling): Do frontier models improve reliably over prior-generation models on CAUSALT3?

RQ4 (Mitigation): Can the diagnosed failures be substantially mitigated at inference time through process verification (RCA)?

4.1 Experiment Setup

Data and label sets. We evaluate on CAUSALT3-Seed (454 expert-curated vignettes). Each Pearl level uses a label set matched to the judgment required. L1 uses YES/NO (is the associational claim causal?). L2 uses VALID/FLAWED (is the interventional reasoning sound?). L3 uses VALID/INVALID/CONDITIONAL (is the counterfactual warranted, or is the scenario underdetermined?). The three-way L3 set enables measurement of calibrated abstention. Appendix A depicts domain breakdown, trap taxonomy, and protocol.

Models. We evaluate a range of frontier and prior-generation models: GPT-4-Turbo, GPT-5.2, Claude 3.5 Sonnet, Claude 3.5 Haiku, and GPT-3.5.² We additionally evaluate an augmented set-

ting where a base model is wrapped with the Recursive Causal Audit (RCA). RCA enforces trace-output consistency through staged escalation and judge-based verification (Section 3; implementation details in Appendix D). This condition tests whether the diagnosed failures can be substantially mitigated at inference time without retraining.

Statistical analysis. All experiments use temperature $T=0$ for reproducibility. We report 95% Clopper-Pearson exact binomial confidence intervals (CIs) for all proportions. Worst-case CI half-widths are $\pm 14.5\%$ at $N=50$ (L1 per-class), $\pm 10.2\%$ at $N=100$ (L1/L3 overall), and $\pm 5.8\%$ at $N=304$ (L2).

For primary between-model comparisons, we report two-proportion z -tests as a simple effect-size significance check. We designate 10 primary comparisons (the headline findings in RQ1 – RQ4) and apply a Bonferroni-adjusted threshold of $\alpha=0.005$. The largest reported effect, the 55-point L3 Safety gap between GPT-4-Turbo ($\hat{p}=0.75$, CI [65.3, 83.1]) and GPT-5.2 ($\hat{p}=0.20$, CI [12.7, 29.2]), yields $z=7.79$, $p < 10^{-14}$, surviving correction by a wide margin. Smaller effects (e.g., L1 Utility differences) should be interpreted with the per-class CI widths in mind; we flag cases where CIs overlap.

4.2 Experimental Results

We report results on CAUSALT3-Seed across Pearl’s hierarchy. Across levels, we evaluate both *Utility* (endorsing valid causal claims, Sheep) and *Safety* (rejecting invalid causal claims, Wolves), and separately measure calibrated abstention on underdetermined cases. Two recurring error profiles emerge: over-endorsement (accepting Wolves) and over-rejection (rejecting Sheep). Their balance varies across L1 through L3 and across protocols.

Table 3: **L1 Association Results.** Safety is uniformly high, but Utility reveals over-refusal in some models.

Model	Util. (Sheep)	Safety (Wolves)	Overall
GPT-4-Turbo	100.0	100.0	100.0
GPT-5.2	90.0	100.0	95.0
GPT-3.5	90.0	100.0	95.0
Claude 3.5 Sonnet	60.0	100.0	80.0
Claude 3.5 Haiku	40.0	96.0	68.0

4.2.1 L1 Association (Spurious Correlation)

Level 1 tests whether models distinguish correlation from causation under purely observational evidence. L1 includes both valid causal conclusions

²All API calls were made between 2025 and January 2026.